Subject: COMPILER DESIGN | **Type:** Previous Year Question Paper
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Mid-Semester Examination I - Question Paper (2025)

Questions & Mark Allocations

1. Explain the various phases of a compiler with a neat block diagram using position = initial + rate * 60. (10 Marks)

2. Construct a DFA for the regular expression (a|b)*abb using Thompson's and subset construction. (10 Marks)

3. Compute FIRST and FOLLOW sets for the given grammar:

E -> T E'

E' -> + T E' | epsilon

T -> F T'

T' -> * F T' | epsilon

F -> (E) | id (10 Marks)

4. Check whether the above grammar is LL(1) by constructing the predictive parsing table. (10 Marks)

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